

calSchedules, Module Reference

How calibration work is planned, grouped and advanced

calSchedules decides *when* each device is calibrated and *with whom*. CalSoft performs the calibration; this module plans it, keeps devices that should be visited together in the same month, and rolls a group forward once it is done.

This document covers what the module contains, how a schedule moves through its life, and, in section 7, what is not yet finished, stated in full rather than left to be discovered.

1. The central idea: groups

A schedule is not planned per device. It is planned per **group**, because a technician visiting a ward calibrates everything in that ward on one trip.

group = (grouping key, scheduled month)

Two grouping keys are supported:

Planning logic	Groups by	Suits
Department	Where the device lives	A visit to one ward
Description	What kind of device it is	One technician, one device type, across wards

Everything else follows from this. A group **moves together**, **completes together**, and **advances together**.

1.1 Two vocabularies, one meaning

The same two concepts carry two sets of names, for historical reasons:

Stored as	Also stored as	Means
date_based	department	Group by department
description_based	description	Group by equipment type

These are normalised when read, so a schedule stamped either way groups the same. No migration was needed and stored values are left alone.

1.2 Who counts as a member

One rule: the equipment and the schedule must both be active.

This matters more than it sounds. Two parts of the system previously disagreed, one required the equipment active and ignored the schedule flag, the other the reverse, so a retired device held its group open in one path and a deleted schedule held it open in the other. Either way the group never read as complete and never advanced. There is now one definition, and it is the default, so a new caller cannot reintroduce the split by omission.

2. The life of a schedule

Status	Meaning
pending	Planned, not yet started
pushed	Deliberately moved to a later month
in_progress	Work has begun
pending_approval	Calibration done, awaiting review
completed	Approved and closed
overdue	Past its due month

2.1 Due dates fall at the end of the month

A calibration is due **within a month**, not on a particular day of it:

$$\text{due} = \text{month-end}(\text{last calibration} + \text{interval})$$

A device calibrated on 3 March with a 12-month interval is due **31 March** the following year, not the 3rd. The workshop therefore has the whole month to fit the visit, and every calibration performed in a month shares one due date which is what makes a group movable.

Intervals are calendar months, not day counts. Adding $30 \times \text{months}$ treats a year as 360 days, so a yearly calibration drifts about five days earlier each cycle and slips into the previous month after roughly six years.

2.2 Completion and advance

When the **last** member of a group completes:

1. the completed schedule is locked immediately;
2. the whole group advances to `scheduled month + interval`, keeping its planning logic;
3. the new schedules are stamped `signal`, marking them as automatically created.

The new month is based on the **scheduled** month, not the actual calibration date, so calibrating early or late does not drag the schedule with it.

3. Regrouping

Two different operations, often confused.

3.1 Moving a selection

Move a chosen set of schedules into another month, a ward being refurbished, or devices that were scheduled apart and should be visited together. Previewable: the preview runs the same code and rolls back, so what you are shown is what will happen, including what will be refused and why.

Alignment stands down during a deliberate move. The system normally enforces "equipment never jump between months within a group", which is right for accidental drift and fatal for an intentional move, moving the first member would snap it back to where the rest still are, so the move could never start.

3.2 Changing the planning logic

Re-normalise every pending schedule when the whole estate switches from department-based to description-based planning, or back. This is an estate-wide operation and a different decision from moving one group.

3.3 What can never be moved

Three protections, applied identically by both operations:

Protected	Why	Remedy
Completed	A finished calibration is history	None. It is a record
Locked	Deliberately frozen	Unlock it first
Automatically created (signal, locker, job_card)	Will be regenerated on the next cycle	Change the cycle, not the row

The reason is always reported, not just the refusal, because the remedy differs.

4. The group view

Groups are shown as cards, because a group is what the scheduler acts on: completion out of total, a progress bar, how many remain, where the group rolls to when it finishes, and per-member state with protected rows marked and explained. The least-covered groups sort first.

5. The parts of the module

Part	Responsibility
grouping	The shared brain: group identity, membership, next period, due dates, certificate-number format. Pure, no database above the one query builder, and unit-tested without one.
regroup	Targeted moves and the group snapshot
instant_reconciliation	Signals: complete on certificate, advance on group completion, align to group
locker	Freezing completed work and rescheduling locked groups
tasks	Batch equivalents for catch-up and periodic runs
reconciliation	Fallback paths when the instant machinery is unavailable
views	Dashboard, schedules, groups, bulk operations, exports

6. Protections in force

Protection	Guards against
Three-layer move protection	Rewriting history or fighting the auto-scheduler
Group alignment	A device drifting out of its group by accident

Protection	Guards against
Immediate locking on completion	A finished calibration being reorganised
Conflict detection on move	Two schedules for one device in one month
Advisory-locked numbering at HQ	Two certificates sharing a number

7. What is not yet finished

Stated in full. Each of these is real, and none is hidden behind a reassuring default.

7.1 Two models for one concept

There are **two** CalibrationSchedule models, one in calSchedules, one in CalSoft. The machine reports read one; calibration sessions link to the other. They now agree on due dates, but they remain two models for one idea, and a future change must be made in both. **Not yet reconciled.**

7.2 Overlapping reschedule mechanisms

Five paths can advance a schedule: the instant signal, the batch task, the locker, the reconciliation fallback, and a manual move. They share the next-period calculation, so they agree on *when*. They have not been consolidated into one path, so they can still differ in *whether* they fire.

7.3 Remaining membership divergences

The canonical rule is applied in the paths that decide completion and alignment. Dashboard counters elsewhere still use their own filters. Those are display totals rather than scheduling decisions, so they cannot block a group but a count on one screen may not match a count on another.

7.4 No department-level scheduling view

Coverage by department is shown on the HOD dashboard. The scheduling screens themselves are workshop-wide, so there is no way to plan one department's calendar in isolation.

7.5 Intervals are limited to 6 or 12 months

The interval field offers two values. A device needing a 3-month or 24-month cycle cannot be represented, and drift-based interval estimation, which the calculation engine can compute, is not wired into planning.

7.6 No capacity planning

Nothing checks whether a month's scheduled work can actually be done by the people available. A reorganisation can place more work in one month than the workshop can absorb, and nothing objects.

7.7 Overdue is computed, not stored

overdue exists as a status but is derived on read from the due date. A schedule sitting in pending past its month is overdue in every display and still pending in the database, so a report written against the stored status will undercount.

Cirqen Calibration Software. The calibration this module plans is documented in CalSoft, Module Reference; the mathematics in The Mathematics of Calibration.